SPETC v10: Physics of the Spectro-Photometric Exposure Time Calculator

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Abstract

SPETC is an exposure time calculator (ETC) for ground-based broad-band photometry and low- to high-resolution spectroscopy, descended from the original Fortran `interpola_spad` chain and rewritten in Python on `astropy`. This document derives every physical relation the code evaluates: the radiometric chain from a CALSPEC-calibrated template spectrum to detected photo-electrons; synthetic photometry in the SVO convention; atmospheric extinction, refraction, airmass and differential dispersion; the position-dependent night-sky model with van Rhijn airglow, zodiacal light, integrated starlight and Krisciunas–Schaefer moonlight; Gaussian and Moffat point-spread functions; the complete noise budget including scintillation and quantization; slit and slitless (Star Analyser) spectroscopy; and the treatment of extended sources. Expected performance and the known limits of validity are summarized.

Contents

1 Introduction

An exposure time calculator answers two reciprocal questions: what signal-to-noise ratio (S/N) a given exposure of a given target will reach, and how long an exposure is needed to reach a requested S/N. SPETC answers both for broad-band aperture photometry and for slit or slitless spectroscopy, for any telescope–detector combination the user describes, at any site, time and sky position.

The code descends from the Fortran `interpola_spad` programs (2007) and retains their template-calibration convention (Sect. 2.3), while the physics has been rebuilt in Python on `astropy` [Astropy Collaboration, 2022]: all radiometric quantities carry explicit units, all coordinate and time transformations use IERS-based ERFA routines, and every empirical relation is taken from the published literature cited in the section where it is used.

Version 10 upgrades the physical model in six areas, each described in its own section below:

1. a position-dependent night-sky model with van Rhijn airglow, zodiacal light, integrated starlight, and the Krisciunas and Schaefer [1991] moonlight model with the published scattering function (Sect. 4);
2. atmospheric refraction and the Pickering [2002] airmass (Sect. 3.2);
3. selectable Gaussian or Moffat [1969] point-spread functions (Sect. 5);
4. a noise budget extended with Young [1967] scintillation, Filippenko [1982] atmospheric dispersion and ADC quantization noise (Sects. 3.3, 6.1);
5. transmission-grating (Star Analyser 100/200) slitless spectroscopy (Sect. 7.2);
6. extended sources: galaxies, nebulae and planets treated as uniform surface brightness (Sect. 6.6).

Throughout, wavelengths are in Ångström internally, fluxes are F_λ in erg s^{-1} cm^{-2} Å^{-1}, and detected signals are in photo-electrons. The notation is summarized in Table 1.

2 Radiometric framework

2.1 Magnitude systems and zero points

A magnitude m in a system with F_ν zero point Z_ν (in Jy) corresponds to the monochromatic flux density

$$F_\lambda(\lambda) \;=\; Z_\nu \, \frac{c}{\lambda^2} \, 10^{-0.4\,m}, \tag{1}$$

Table 1: Principal symbols.

Symbol	Meaning
F_λ	spectral flux density [erg s^{-1} cm^{-2} Å^{-1}]
A	unobstructed collecting area $\pi(D^2 - d^2)/4$
T_λ	filter transmission (0–1)
Q_λ	detector quantum efficiency (0–1)
η	scalar optics throughput excluding QE
O_λ	optional calibrated instrument response curve
$T_{\text{atm}}(\lambda)$	zenith atmospheric transmission
X	airmass
z	zenith distance; h apparent altitude
s	seeing FWHM [arcsec]
q	parallactic angle
S, B, D	source, sky, dark electrons in the aperture
σ_R	read noise per pixel [e$^-$ rms]
g	gain [e$^-$/ADU]
N_{pix}	pixels in the photometric aperture or extraction
R	resolving power $\lambda/\Delta\lambda$

which the code evaluates with `astropy` units (`magnitude_f_lambda`). The AB system [Oke and Gunn, 1983] uses $Z_\nu = 3631$ Jy at every wavelength; Vega systems use the per-filter zero points delivered inside each SVO filter profile [Rodrigo and Solano, 2020, Bessell et al., 1998].

2.2 Synthetic photometry (SVO convention)

The synthetic magnitude of a spectrum F_λ in a filter with response T_λ is

$$m = -2.5 \log_{10} \frac{\int F_\lambda T_\lambda W_\lambda \, d\lambda}{\int F_\lambda^0 T_\lambda W_\lambda \, d\lambda}, \qquad W_\lambda = \begin{cases} \lambda & \text{photon counter} \\ 1 & \text{energy integrator,} \end{cases} \tag{2}$$

where F_λ^0 is the zero-magnitude spectrum of Eq. (1) and the weighting follows the SVO `DetectorType` flag of the profile. Both conventions are implemented and regression-tested; for a coloured source they differ measurably.

2.3 Template calibration

Template spectra are absolutely calibrated CALSPEC/BPGS spectrophotometry [Bohlin et al., 2014]; the catalogue field m_{v0} records the visual magnitude represented by each file. The code first restores the visual-zero distribution, $F_\lambda \to F_\lambda \, 10^{0.4 m_{v0}}$ (the original Fortran convention), then scales it to the user's reference measurement:

$$F_\lambda^{\text{cal}} = F_\lambda \, 10^{0.4 m_{v0}} \, 10^{-0.4 \left[m_{\text{ref}} - (m_{\text{ref}}^{\text{syn}} - m_V^{\text{syn}})\right]}, \tag{3}$$

where the synthetic colour term $(m_{\text{ref}}^{\text{syn}} - m_V^{\text{syn}})$ is computed from the template itself with Eq. (2) whenever the reference filter differs from the visual response, and is exactly zero in the original V-only case, which then reduces to the Fortran scale factor $10^{-0.4(V_{\text{target}} - m_{v0})}$.

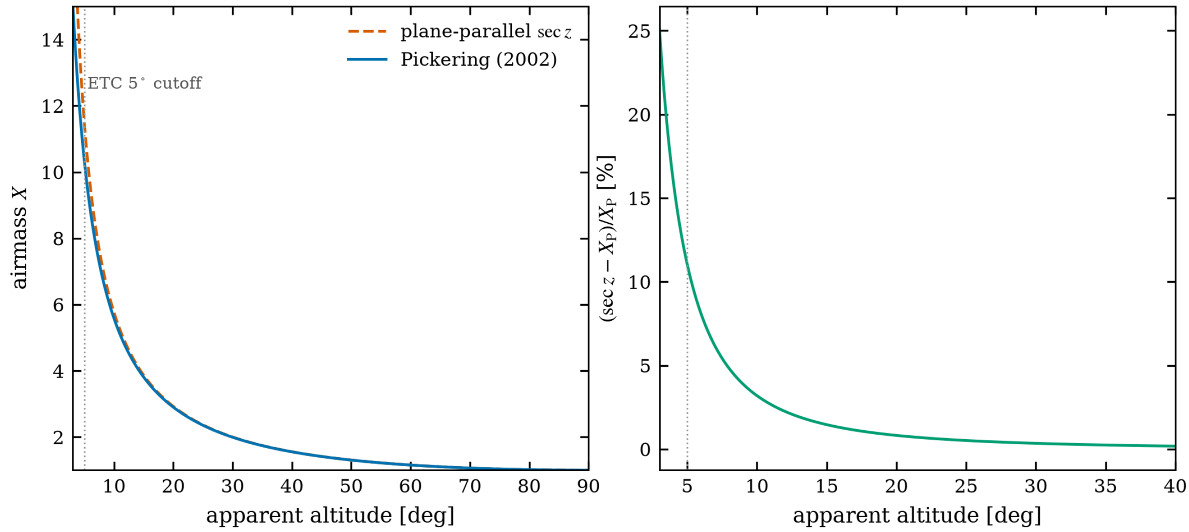

Figure 1: Left: Pickering (2002) airmass versus plane-parallel sec z. Right: relative overestimate of sec z; at the ETC's $5°$ cutoff it already exceeds 10%.

2.4 Detected electron rate

The photo-electron rate through the full system is

$$\dot{S} \;=\; A\,\eta \int F_\lambda^{\rm cal}\, T_\lambda\, Q_\lambda\, O_\lambda\, T_{\rm atm}(\lambda)^X\, \frac{\lambda}{hc}\, d\lambda, \tag{4}$$

with the trapezoidal quadrature carried out on the filter grid (photometry) or per resolution element (spectroscopy). Wavelength coverage is a hard requirement: a template, QE curve, filter or atmospheric table that does not cover the requested interval raises an error instead of silently zero-filling.

3 Atmosphere

3.1 Extinction

Zenith transmission $T_{\rm atm}(\lambda)$ comes either from a site-specific FITS table (e.g. the Paranal product, whose `EXTINCTION` column in mag airmass^{-1} is converted once as $T = 10^{-0.4k_\lambda}$) or from the built-in broad-band U–K table (Table 2). The Bouguer law applies it as T^X, exactly once, to the source; observed sky brightnesses are never extinguished a second time.

3.2 Refraction and airmass

Apparent altitudes include ERFA atmospheric refraction evaluated at the ISA pressure and temperature of the site elevation, $P = 1013.25\,\mathrm{hPa}\, e^{-E/8435\,\mathrm{m}}$ and $T = 15\,°\mathrm{C} - 0.0065\,E$. The airmass is the Pickering [2002] interpolative formula on the *apparent* altitude h (degrees),

$$X \;=\; \frac{1}{\sin\!\big(h + 244/(165 + 47\,h^{1.1})\big)}, \tag{5}$$

which matches sec z near the zenith and remains physical to the horizon, where the plane-parallel form diverges (Fig. 1). The ETC deliberately reports no airmass below $h = 5°$.

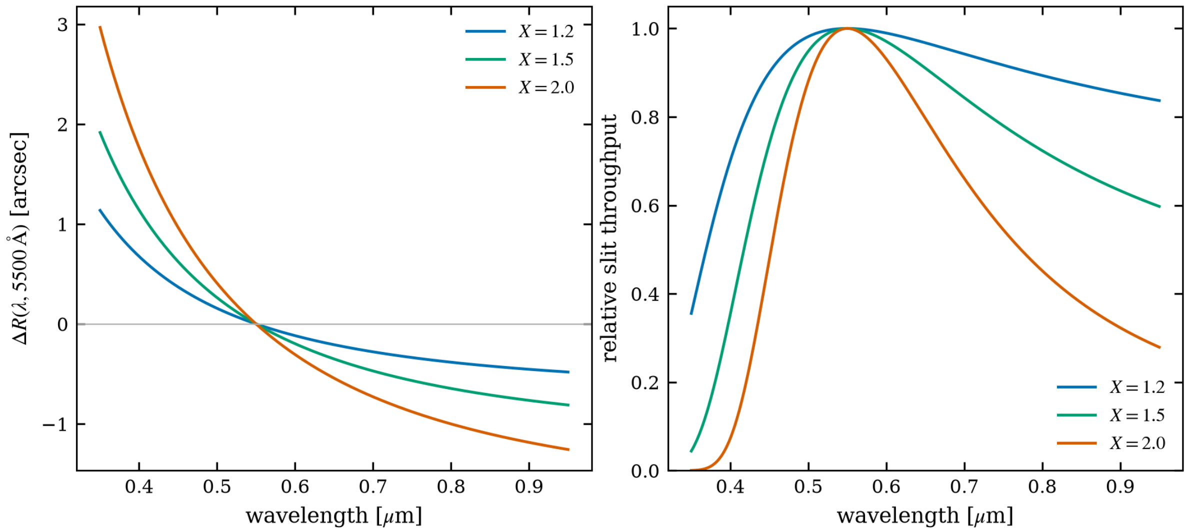

Figure 2: Left: differential refraction relative to 5500 Å for three airmasses. Right: resulting relative slit throughput for a $1.5''$ slit in $1.5''$ seeing when the slit is *not* at the parallactic angle (worst-case geometry).

3.3 Differential atmospheric refraction

The refractive index of air follows Filippenko [1982]: at standard conditions

$$(n-1)\,10^6 = 64.328 + \frac{29498.1}{146-\sigma^2} + \frac{255.4}{41-\sigma^2}, \qquad \sigma = 1/\lambda_{\mu m}, \tag{6}$$

rescaled to the site pressure and temperature with his equations (2)–(3), including the water-vapour term. The differential refraction relative to a reference wavelength is

$$\Delta R(\lambda) = 206265\,\big[n(\lambda) - n(\lambda_{\text{ref}})\big]\,\tan z, \tag{7}$$

about $1''$ between 4000 and 5500 Å at $z = 45°$ (Fig. 2). Its use in slit losses is described in Sect. 7.1; the parallactic angle

$$q \;=\; \arctan\frac{\sin H}{\tan\varphi\cos\delta - \sin\delta\cos H} \tag{8}$$

(H hour angle, φ latitude, δ declination) is reported for every time sample, because a slit oriented at q suffers no dispersion loss.

3.4 Scintillation

Intensity scintillation follows the Young [1967] law, kept in the functional form endorsed by Osborn et al. [2015],

$$\frac{\sigma_{\text{scint}}}{I} \;=\; 0.09\,d_{\text{cm}}^{-2/3}\,X^{1.75}\,e^{-E/8000\,\text{m}}\,(2\,t_{\text{exp}})^{-1/2}, \tag{9}$$

with d the aperture diameter, E the site elevation and t_{exp} the exposure time. Because this noise is multiplicative in the source flux, it dominates bright-star photometry and imposes a hard S/N ceiling $\simeq I/\sigma_{\text{scint}}$ per exposure (Fig. 6).

4 Night-, twilight- and day-sky model

4.1 Dark sky

The dark-sky normalization is the broad-band U–K surface-brightness table of Table 2 (from the ING model of Benn and Ellison 1998), corrected for the pointing and epoch through the S10 component budget

$$q_3 \;=\; q_{\text{air}}\,V(z) \;+\; q_{\text{zod}}(|\beta|) \;+\; q_*(|b|), \tag{10}$$

Table 2: Built-in broad-band sky data (U–K): dark-sky surface brightness, extinction coefficient, Vega zero point and Krisciunas–Schaefer lunar colour factor.

Band	λ [Å]	$\mu_{\rm dark}$ [mag/$''^2$]	k [mag/X]	$Z_\nu^{\rm Vega}$ [Jy]	lunar factor
U	3650	22.0	0.550	1810	2.00
B	4450	22.7	0.250	4260	1.60
V	5510	21.9	0.150	3640	2.40
R	6580	21.0	0.090	3080	3.00
I	8060	20.0	0.060	2550	5.90
Z	10000	18.8	0.050	2250	8.94
J	12500	16.1	0.100	1594	13.06
H	16500	14.7	0.110	1024	18.09
K	22000	12.5	0.070	667	24.04

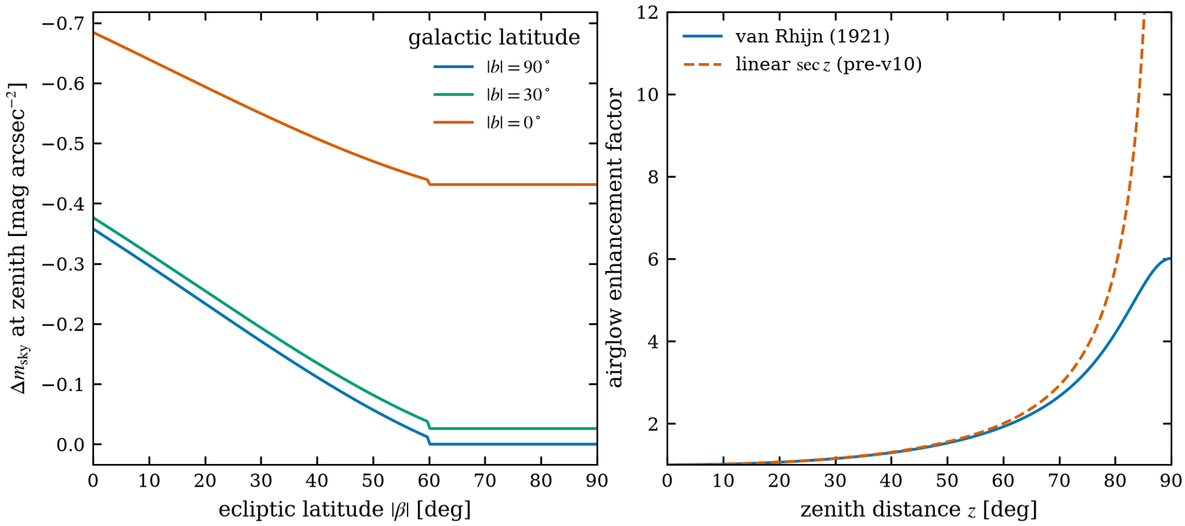

Figure 3: Left: zenith dark-sky brightening relative to the tabulated values as a function of ecliptic latitude, for three galactic latitudes (solar minimum). Right: van Rhijn airglow enhancement versus the linear sec z scaling used before v10.

normalized to the dark reference composition $q_{\rm ref} = 145 + 60$ S10. The components are:

- **Airglow**: $q_{\rm air} = 145 + 130\,(a - 0.8)/1.2$ S10, where the solar-cycle activity $a \in [0.8, 2.0]$ is interpolated between the tabulated solar minima. The line-of-sight enhancement is the van Rhijn path-length law [van Rhijn, 1921] for an emitting layer at $\hat{h} = 90$ km,

$$V(z) = \left[1 - \left(\frac{R_\oplus}{R_\oplus + \hat{h}} \sin z\right)^2\right]^{-1/2}, \tag{11}$$

  which is sub-linear in airmass and bounded ($V \simeq 6$) at the horizon (Fig. 3, right).

- **Zodiacal light**: $q_{\rm zod} = 140 - 90 \sin|\beta|$ S10 for $|\beta| < 60°$, else 60 S10 — symmetric about the ecliptic by construction.

- **Integrated starlight**: $q_* = 100\, e^{-|b|/10°}$ S10.

The field's ecliptic and galactic latitudes are computed from the entered ICRS coordinates with `astropy`; the sky is therefore genuinely position-dependent (Fig. 3, left).

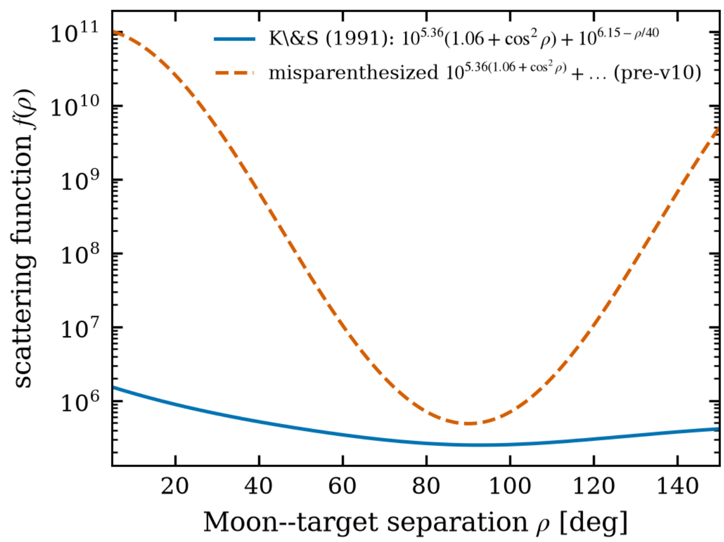

Figure 4: The Krisciunas and Schaefer [1991] scattering function (solid) and the misparenthesized variant present before v10 (dashed).

4.2 Moonlight

Moonlight follows Krisciunas and Schaefer [1991]. The lunar illuminance for phase angle α (0 = full Moon) is $I^* = 10^{-0.4(3.84+0.026|\alpha|+4\times10^{-9}\alpha^4)}$, and the scattering function at Moon–target separation ρ is

$$f(\rho) \;=\; 10^{5.36}\big(1.06+\cos^2\rho\big) \;+\; 10^{6.15-\rho/40^\circ}. \tag{12}$$

The surface brightness in nanoLambert is $B_{\rm moon} = I^* f(\rho)\,10^{-0.4kX_{\rm m}}\big(1-10^{-0.4kX}\big)$, converted to S10 and distributed over the nine bands with the lunar colour factors of Table 2. Equation (12) corrects a long-standing transcription error in which the constant $10^{5.36}$ appeared inside the exponent; the two forms diverge by more than three orders of magnitude within 30° of the Moon (Fig. 4).

4.3 SQM and Bortle input

Amateur observers know their sky as a Sky Quality Meter reading (zenith V surface brightness) or a Bortle [2001] class rather than as a per-band table. In `sqm` mode the entered zenith SQM value $\mu_{\rm SQM}$ replaces the dark-sky normalization directly:

$$\mu(\lambda_{\rm piv}) \;=\; \mu_{\rm SQM} \;+\; \big[\mu_{\rm tab}(\lambda_{\rm piv}) - \mu_{\rm tab}(V)\big], \tag{13}$$

i.e. the built-in table supplies only the band colour, while the SQM reading — which already contains the site's light pollution and current airglow — sets the level. The Krisciunas–Schaefer Moon contribution of the previous subsection is applied on top unchanged. A Bortle-class selector fills typical zenith SQM values (21.9, 21.8, 21.6, 21.1, 20.5, 19.8, 19.0, 18.0, 17.0 mag arcsec^{-2} for classes 1–9).

4.4 Twilight and daylight

Between Sun altitudes of -18° and 0° the dark and daylight skies are blended linearly in flux. Above the horizon the V-band daylight surface brightness of Weaver [1947] is used, mapped to other bands with the dark-sky colour of Table 2. Both regimes are broad-band approximations, clearly flagged as such in the user interface.

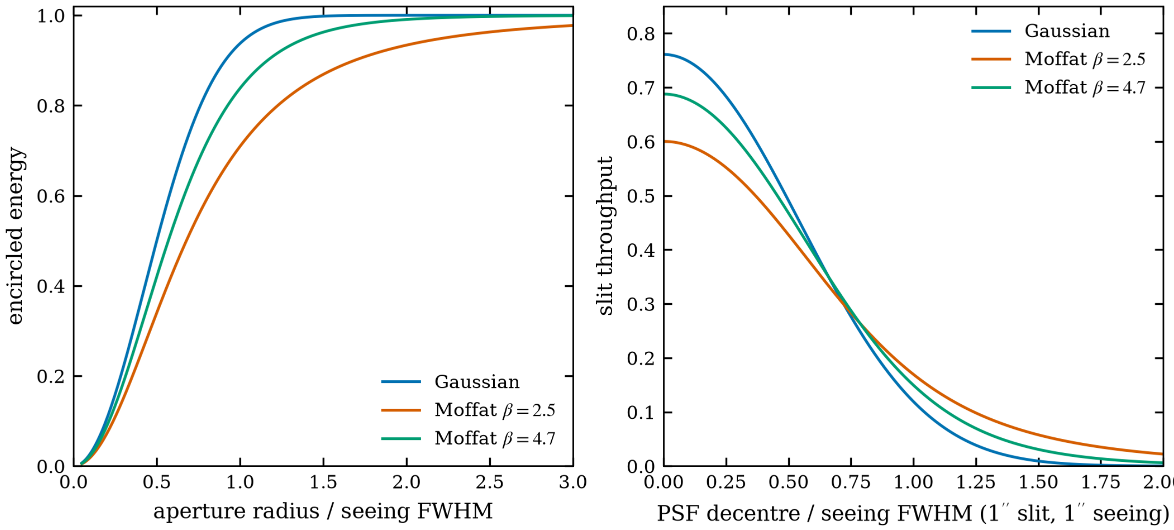

Figure 5: Left: encircled energy versus aperture radius in units of the seeing FWHM. Right: slit throughput of a slit of one seeing-FWHM width as the PSF is decentred, Eq. (16); the Moffat wings both lower the centred coupling and soften the decentring penalty.

5 Point-spread function and aperture coupling

The seeing-limited PSF is selectable between a circular Gaussian and a circular Moffat [1969] profile

$$I(r) \propto \left(1 + \frac{r^2}{\alpha^2}\right)^{-\beta}, \qquad \alpha = \frac{\mathrm{FWHM}}{2\sqrt{2^{1/\beta} - 1}}, \tag{14}$$

whose power-law wings ($\beta \simeq 2.5$–4.7; the Gaussian is the $\beta \to \infty$ limit) reproduce real seeing profiles that a Gaussian underestimates.

Encircled energy. For a circular aperture of radius r,

$$\mathrm{EE}_{\mathrm{Gauss}}(r) = 1 - e^{-r^2/2\sigma^2}, \qquad \mathrm{EE}_{\mathrm{Moffat}}(r) = 1 - \left(1 + \frac{r^2}{\alpha^2}\right)^{1-\beta}. \tag{15}$$

Slit coupling. The flux through an infinite slit of width w whose centre is displaced by δ from the PSF centroid uses the one-dimensional marginal profile. For the Gaussian this is the usual error function; for the circular Moffat the marginal is *exactly* a Student-t distribution with $\nu = 2\beta - 2$ degrees of freedom and scale $\alpha/\sqrt{\nu}$, so

$$T(w, \delta) \;=\; F_\nu\!\left(\frac{w/2 - \delta}{\alpha/\sqrt{\nu}}\right) - F_\nu\!\left(\frac{-w/2 - \delta}{\alpha/\sqrt{\nu}}\right), \tag{16}$$

with F_ν the Student-t CDF — no numerical integration is required. The same expression with w equal to one pixel gives the central-pixel fraction used for saturation prediction, as the separable product of the dispersion and spatial couplings. Figure 5 compares the models.

6 Photometric ETC

6.1 Signal and noise budget

For a circular aperture of radius r containing $N_{\mathrm{pix}} = \pi r^2/p^2$ pixels (p = plate scale per pixel), the extracted source signal is $S = \dot{S}\,\mathrm{EE}(r)\,t_{\mathrm{exp}}$, the sky signal is $B = \dot{B}\,\pi r^2\,t_{\mathrm{exp}}$ from the observed surface brightness

Table 3: Noise terms and their scalings.

Term	Variance [e^{-2}]	Scaling	Reference
Source shot	S	$\propto t$	—
Sky shot	B	$\propto t\, r^2$	Sect. 4
Dark	D	$\propto t\, N_{\text{pix}}$	—
Read	$N_{\text{pix}} \sigma_R^2$	per frame	—
Scintillation	$(S\, \sigma_{\text{scint}}/I)^2$	$\propto t^{1\ a}$	Young [1967], Osborn et al. [2015]
Quantization	$N_{\text{pix}}\, g^2/12$	per frame	Janesick [2001]

[a] $S \propto t$ while $\sigma_{\text{scint}}/I \propto t^{-1/2}$, so the S/N ceiling $I/\sigma_{\text{scint}} \propto \sqrt{t}$.

of Sect. 4, and the dark charge is $D = \dot{d}\, N_{\text{pix}}\, t_{\text{exp}}$. The S/N is the CCD equation extended with the multiplicative and quantization terms:

$$\text{S/N} \;=\; \frac{S}{\sqrt{\,S + B + D + N_{\text{pix}}\, \sigma_R^2 + \left(S\, \sigma_{\text{scint}}/I\right)^2 + N_{\text{pix}}\, g^2/12}}, \tag{17}$$

with the scintillation fraction of Eq. (9) and the ADC quantization variance $g^2/12$ per pixel [Janesick, 2001]. Flat-field residuals are deliberately not modelled. The inverse problem — the exposure that reaches a requested S/N — is solved in closed form from the quadratic obtained from Eq. (17) with the Poisson terms proportional to t.

6.2 Exposure solving and stack planning

The exposure reaching a requested S/N solves the quadratic obtained from Eq. (17). Crucially, the scintillation variance is *linear* in time — with $\sigma/I = C(2t)^{-1/2}$ the variance of $\dot{S}t$ electrons is $(\dot{S}C)^2 t/2$ — so it enters the closed form exactly as an additional Poisson-like rate added to $Q = \dot{S} + \dot{B} + \dot{D}$:

$$t \;=\; \frac{\text{SNR}^2\left(Q + \dot{S}^2 C^2/2\right) + \sqrt{\,\cdot\;^2 + 4\dot{S}^2\, \text{SNR}^2 N_{\text{pix}} \sigma_R^2}}{2\dot{S}^2}. \tag{18}$$

Without this term a bright-star exposure request undershoots badly (a demonstration case returned S/N 230 for a 500 target); with it the solved exposure reproduces the requested S/N to better than 2% (regression tested).

The **stack planner** answers the practical amateur question "how many subs?": the sub-exposure is the shortest of the user's preference and the saturation limit of the brightest pixel; the total time for N frames of length t_{sub} follows in closed form from the stacked CCD equation $\text{SNR} = \dot{S}T/\sqrt{QT + N N_{\text{pix}} \sigma_R^2}$ (the scintillation rate is framing-independent, since it averages down with total time). Reported alongside are the read-noise penalty relative to one ideal long exposure and the frame length above which the background variance exceeds $10\, \sigma_R^2$ per pixel — the classical "sky-limited" criterion.

6.3 Differential photometry

For variable-star and exoplanet work the scientific quantity is the differential precision against a comparison star of magnitude m_c, in millimagnitudes per frame:

$$\sigma_{\text{diff}} \;=\; 1085.7\, \sqrt{\text{SNR}_t^{-2} + \text{SNR}_c^{-2}} \quad [\text{mmag}], \tag{19}$$

where both S/N values carry the full budget of Eq. (17) — each star scintillates — and the two scintillation terms are treated as uncorrelated, which is conservative for the arcminute-scale separations typical of ensemble photometry [Osborn et al., 2015].

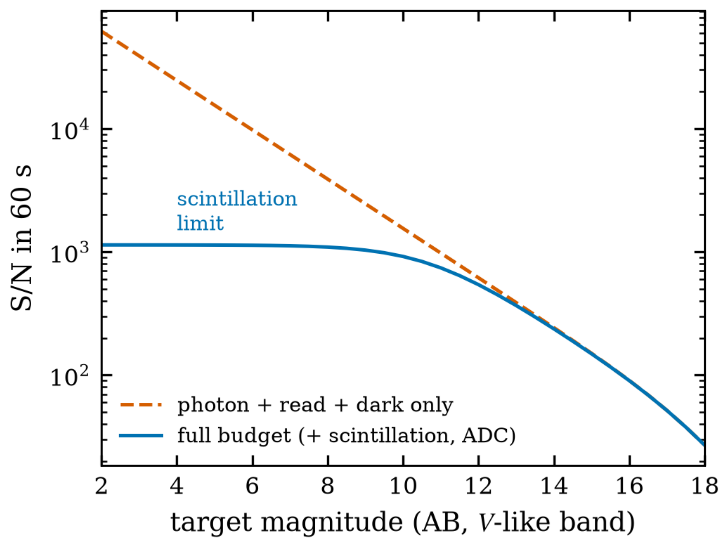

Figure 6: Broad-band S/N in 60 s for the 358-mm reference configuration, computed by the released code. The full budget (solid) saturates at the scintillation ceiling $\simeq 1.2 \times 10^3$; a Poisson-only model (dashed) overestimates bright-star performance by more than an order of magnitude.

6.4 CMOS gain description

Amateur CMOS cameras change conversion gain, read noise and full well *together* with the gain setting. The detector is therefore describable by a per-camera table (setting $\to$ e$^-$/ADU, σ_R, FWC) rather than one fixed triple; the selected row feeds Eq. (17), the quantization term $g^2/12$ and the saturation limits simultaneously, so switching the gain setting consistently moves the S/N, the ADC ceiling and the maximum unsaturated exposure.

6.5 Saturation

Saturation is evaluated on the brightest pixel of the *unextracted* image: the PSF central-pixel fraction (Sect. 5) times the total source charge, plus the per-pixel sky and dark. The peak is compared with both the full well and the ADC ceiling $g\,(2^{N_{\rm bit}} - 1)$, and the flag distinguishes `FULL_WELL`, `ADC`, `BOTH` and `NONE`; the maximum unsaturated single-frame exposure is reported at every time sample.

6.6 Extended sources

Galaxies, nebulae and planets are modelled as uniform surface brightness over a stated angular area Ω: the entered magnitude remains the *integrated* magnitude, so the mean surface rate is $\dot{S}/\Omega$ per arcsec^2. The aperture then captures the area fraction $\min(\pi r^2/\Omega,\, 1)$ with no PSF concentration loss, and the peak pixel holds $\dot{S}\,p^2/\Omega$ — the appropriate saturation criterion for a resolved source. The approximation is stated to be valid when the source is much larger than the seeing disc, where PSF blur merely redistributes light across the source edge.

7 Spectroscopic ETC

All spectroscopic quantities are per resolution element. The wavelength grid is logarithmic at constant R in slit mode, or linear with the resolution element $\Delta\lambda = \mathcal{D}\,\ell$ in slitless mode ($\mathcal{D}$ = dispersion in Å/pixel, ℓ = LSF FWHM in pixels). Source and sky rates follow Eq. (4) restricted to each element, and the S/N applies Eq. (17) with $N_{\rm pix}$ = (dispersion pixels per element) $\times$ (spatial extraction pixels).

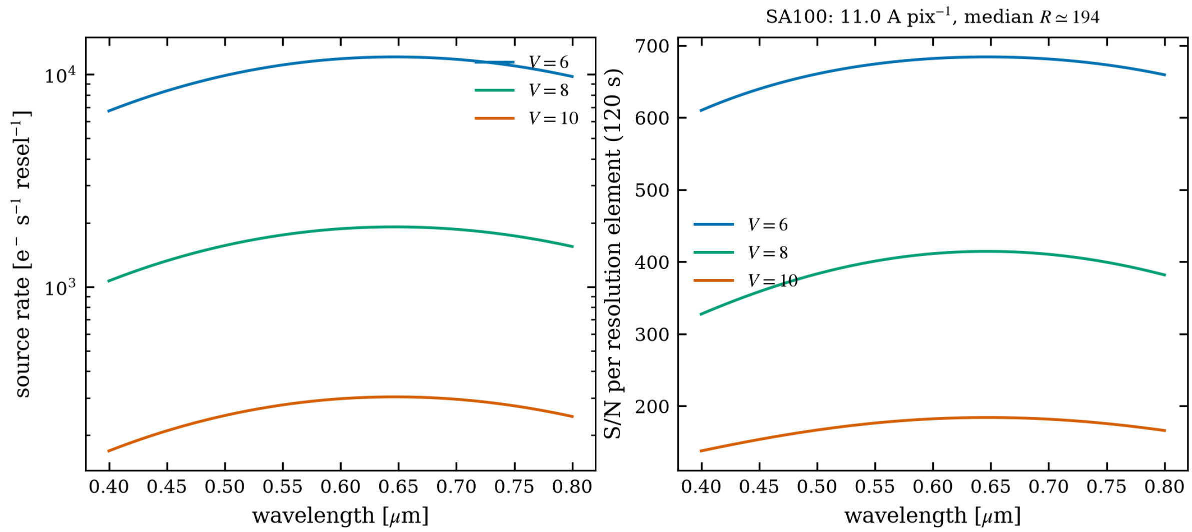

Figure 7: Star Analyser 100 predictions from the released code for a 200-mm f/5 telescope with a typical CMOS camera: detected source rate (left) and S/N per resolution element in 120 s (right) for $V = 6, 8, 10$ flat-spectrum targets.

7.1 Slit mode

The extracted fraction is the product of two couplings of Eq. (16): the slit width across dispersion and the finite extraction height along it. When the slit is *not* at the parallactic angle of Eq. (8), the differential refraction $\Delta R(\lambda)$ of Eq. (7) decentres the PSF in the slit per wavelength; SPETC applies the worst-case geometry (full offset perpendicular to the slit), reproducing the classical Filippenko [1982] blue-end losses (Fig. 2). A measured slit-width/resolving-power curve, when supplied, overrides the nominal R.

7.2 Slitless mode and the Star Analyser

In slitless spectroscopy the resolution element is set by the image quality, not by a slit: $\ell = \sqrt{(s/p)^2 + \ell_0^2 + \ell_{\rm AD}^2}$ combines seeing, the intrinsic grating LSF and the atmospheric-dispersion smear within one element. The sole extraction aperture is the cross-dispersion height; the sky per element covers (dispersion pixels $\times$ extraction height).

For a transmission grating in the converging beam — the Star Analyser 100/200 geometry — the first-order deviation is $x \simeq L\, m\, n\, \lambda$, so the dispersion at the detector is

$$\mathcal{D} \;=\; \frac{10^4\, p_{\mu m}}{L_{\rm mm}\; n_{\rm mm}\; m} \quad [\text{Å/pixel}], \tag{20}$$

with n the groove density (100 or 200 mm^{-1}), L the grating-to-sensor distance and $m = 1$. An SA100 at $L = 42$ mm on 4.63 µm pixels gives $\mathcal{D} = 11.0$ Å/pixel, matching its published calibration; an SA200 at the same distance gives half that. An optional scalar first-order grating efficiency multiplies source and sky alike. Under $2.5''$ seeing the effective resolving power is $R \simeq 100$–200 (Fig. 7), which is why the mode remains flagged *experimental* until validated against a calibrated instrument.

7.3 Slit-spectrograph geometry

Amateurs know their spectrograph by its hardware — groove density n, collimator and camera focal lengths $f_{\rm coll}$, $f_{\rm cam}$, slit width — not by R. In the Littrow approximation the grating equation gives $\sin\beta = mn\lambda/2$ and a reciprocal linear dispersion $d\lambda/dx = \cos\beta \cdot 10^7/(m\, n\, f_{\rm cam})$ [Å/mm]. The resolution element on the detector is the narrower of the geometric slit image and the seeing image, both reimaged by $f_{\rm cam}/f_{\rm coll}$

(anamorphism 1 in Littrow):

$$\Delta\lambda = \min(w_{\text{slit}}, w_{\text{seeing}})\,\frac{f_{\text{cam}}}{f_{\text{coll}}}\,\frac{d\lambda}{dx}, \qquad R = \frac{\lambda}{\Delta\lambda}, \tag{21}$$

with the widths converted from arcsec at the telescope focal plane. The calculator reports whether the configuration is slit- or seeing-limited and feeds the ETC's resolving-power field, which remains editable; an LHIRES III-like configuration (2400 lines/mm, $f = 200$ mm Littrow, 25-µm slit) returns $R \simeq 2 \times 10^4$ at Hα, in line with the instrument's specification.

7.4 Equivalent-width detectability

The spectroscopic S/N is turned into a science answer through the Cayrel [1988] relation for the equivalent-width uncertainty of a line of FWHM W sampled with pixels of width δx:

$$\sigma(\text{EW}) \;\simeq\; \frac{1.5\,\sqrt{W\,\delta x}}{(\text{S/N})_{\text{pixel}}}, \tag{22}$$

evaluated per wavelength with the resolution element as W and reported in mÅ in every result table. The practical criterion: a line is securely measurable when its expected equivalent width exceeds $\simeq 3\,\sigma(\text{EW})$ — e.g. $\sigma(\text{EW}) = 15$ mÅ means Hα variations of a Be star at the 50 mÅ level are detectable, while 20 mÅ photospheric lines are marginal.

7.5 Time series and reverse solving

Planning series evaluate the S/N reference wavelength bin at every time sample of the visibility window (full spectra only at the selected time and the plot-slider samples), with the sky model, airmass, refraction and parallactic angle recomputed per sample. Target-S/N requests use the closed-form solver of Sect. 6.1 per element and are checked against the saturation limit rather than being silently declared feasible.

8 Filter profiles and synthetic calibration

Professional passbands (Johnson–Cousins, Sloan, 2MASS, ...) ship as SVO Filter Profile Service VOTables [Rodrigo and Solano, 2020]. Amateur filters (Astrodon, Astronomik, Baader, Optolong, Chroma RGB and narrowband sets) are absent from SVO, but a measured or digitized transmission curve is all that is required: the zero points are *computed*, not measured. The bundled `make_filter_profile.py` applies the same synthetic convention SVO itself uses:

- **AB**: $Z_\nu = 3631$ Jy at every wavelength by definition [Oke and Gunn, 1983];
- **Vega**: the photon-weighted mean flux density of the CALSPEC Vega spectrum [Bohlin et al., 2014] through the filter,

$$Z_\nu \;=\; \frac{\displaystyle\int F_\lambda^{\text{Vega}}\,T_\lambda\,\lambda\,d\lambda}{\displaystyle\int \frac{c}{\lambda^2}\,T_\lambda\,\lambda\,d\lambda}, \tag{23}$$

  i.e. the flux a magnitude-zero star delivers in this band.

The tool writes full SVO-format VOTables with the complete photdm annotation and every characterising quantity measured from the curve by the SVO definitions (mean, pivot, effective and photon-weighted wavelengths, 1%-of-peak limits, effective width, FWHM, and the mean solar flux `Fsun` from the CALSPEC solar spectrum), so the produced files are drop-in interchangeable with SVO downloads and land on exactly the same photometric scale as the shipped professional profiles.

Table 4: Validation of the synthetic calibration against the shipped SVO 2MASS.H profile (whose declared values are independent of this code).

Quantity	SVO declared	Computed here	Difference
Pivot wavelength [Å]	16494.9	16494.7	-0.00%
Photon-weighted wavelength [Å]	16460.6	16423.8	-0.2%
Central wavelength [Å]	16506.1	16487.3	-0.1%
$F_\odot$ [$\mathrm{erg\,cm^{-2}\,s^{-1}\,Å^{-1}}$]	22.44	22.33	-0.5%
Vega zero point [Jy]	1024.0	1030.4	$+0.6\%$

Table 4 validates the pipeline end-to-end on 2MASS.H: the sub-percent agreement (the residuals reflect SVO's slightly different Vega SED normalization) means a profile built from a digitized Astrodon or Astronomik curve is calibrated to the same standard as a professional filter — no laboratory measurement beyond the transmission curve itself is needed.

9 Expected performance and limits of validity

9.1 What to expect

- **Bright stars are scintillation-limited.** For the 358-mm reference telescope, a $V \simeq 5$ star in 60 s reaches $\mathrm{S/N} \simeq 1.3 \times 10^3$, not the $\sim 2 \times 10^4$ a photon-only budget would claim (Fig. 6); the ceiling scales as $d^{2/3} X^{-1.75} \sqrt{t}$.
- **Faint limits are sky-limited.** Beyond $m \gtrsim \mu_{\text{sky}} - 2.5 \log_{10}(\pi r^2)$ the S/N falls as $10^{-0.4m}$ and doubling the exposure gains only $\sqrt{2}$.
- **Moonlight matters within $\sim 90°$.** Full Moon at 30° separation brightens the V sky by $\simeq$ 4–5 mag (Sect. 4), turning background-limited exposures from minutes into hours.
- **Slit spectroscopy away from the parallactic angle** loses tens of per cent at the blue end already at $X = 1.5$–2 (Fig. 2); the Results window reports q at every sample so the slit can be set correctly.
- **Star Analyser work** is photometrically generous (no slit losses) but limited to $R \sim 100$–200 by seeing; S/N per element of several hundred in minutes is realistic for $V \lesssim 8$ (Fig. 7).

9.2 Known approximations

The model is explicit about its accuracy floor: the U–K sky table and its colour interpolation are broad-band, and the twilight/daylight bridge is approximate; the extinction fallback table is a smooth broad-band baseline until a site-specific curve is loaded; scintillation uses the classical Young coefficient (site-specific coefficients from Osborn et al. 2015 can be substituted); atmospheric dispersion in slit mode assumes the worst-case orientation; extended sources are uniform discs; and slitless mode awaits validation against a calibrated instrument. Flat-field and fringing residuals, cosmic rays and sky-subtraction penalties are not modelled.

9.3 Numerical verification

Two test programs ship with the code: a synthetic smoke test that pins the scintillation-limited photometric S/N and the per-element spectroscopic S/N of the reference configuration, and a regression suite that verifies, among others, Eq. (12) against the published scattering function, the ecliptic symmetry of the zodiacal term, Eq. (5) against sec z, the Student-t Moffat coupling, the Filippenko dispersion scale, Eq. (20) for the SA100/SA200, and the extended-source area fractions.

References

Astropy Collaboration. The astropy project: Sustaining and growing a community-oriented open-source project. *ApJ*, 935:167, 2022. doi: 10.3847/1538-4357/ac7c74.

C. R. Benn and S. L. Ellison. La palma night-sky brightness. Technical Report La Palma Technical Note 115, Isaac Newton Group, 1998.

M. S. Bessell, F. Castelli, and B. Plez. Model atmospheres broad-band colors, bolometric corrections and temperature calibrations for o–m stars. *A&A*, 333:231–250, 1998.

R. C. Bohlin, K. D. Gordon, and P.-E. Tremblay. Techniques and review of absolute flux calibration from the ultraviolet to the mid-infrared. *PASP*, 126:711–732, 2014. doi: 10.1086/677655.

J. E. Bortle. Introducing the bortle dark-sky scale. *Sky & Telescope*, 101(2):126–129, 2001.

R. Cayrel. Data analysis. In G. Cayrel de Strobel and M. Spite, editors, *The Impact of Very High S/N Spectroscopy on Stellar Physics, IAU Symposium 132*, pages 345–353, Dordrecht, 1988. Kluwer.

A. V. Filippenko. The importance of atmospheric differential refraction in spectrophotometry. *PASP*, 94: 715–721, 1982. doi: 10.1086/131052.

J. R. Janesick. *Scientific Charge-Coupled Devices*. SPIE Press, Bellingham, 2001.

K. Krisciunas and B. E. Schaefer. A model of the brightness of moonlight. *PASP*, 103:1033–1039, 1991. doi: 10.1086/132921.

A. F. J. Moffat. A theoretical investigation of focal stellar images in the photographic emulsion and application to photographic photometry. *A&A*, 3:455–461, 1969.

J. B. Oke and J. E. Gunn. Secondary standard stars for absolute spectrophotometry. *ApJ*, 266:713–717, 1983. doi: 10.1086/160817.

J. Osborn, D. Föhring, V. S. Dhillon, and R. W. Wilson. Atmospheric scintillation in astronomical photometry. *MNRAS*, 452:1707–1716, 2015. doi: 10.1093/mnras/stv1400.

K. A. Pickering. The southern limits of the ancient star catalog and the commentary of hipparchos. *DIO*, 12:3–27, 2002.

C. Rodrigo and E. Solano. The svo filter profile service. XIV.0 Scientific Meeting of the Spanish Astronomical Society, 2020.

P. J. van Rhijn. On the brightness of the sky at night and the total amount of starlight. *Publications of the Kapteyn Astronomical Laboratory Groningen*, 31:1–83, 1921.

H. F. Weaver. The visibility of stars without optical aid. *PASP*, 59:232–243, 1947. doi: 10.1086/125956.

A. T. Young. Photometric error analysis. vi. confirmation of reiger's theory of scintillation. *AJ*, 72:747, 1967. doi: 10.1086/110303.